When a correlation metric cannot tell learning from non-learning in a spiking sensorimotor network

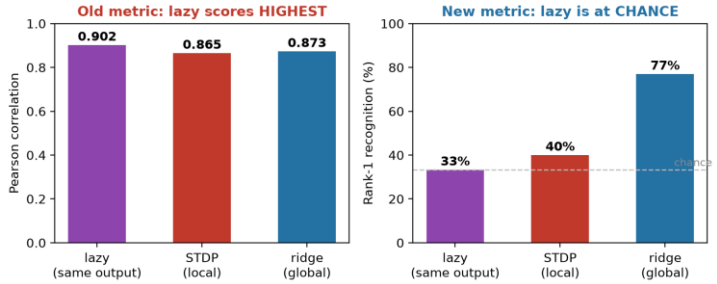
High correlation does not guarantee correct mapping.

Pan Haimeng · SNUFA 2026

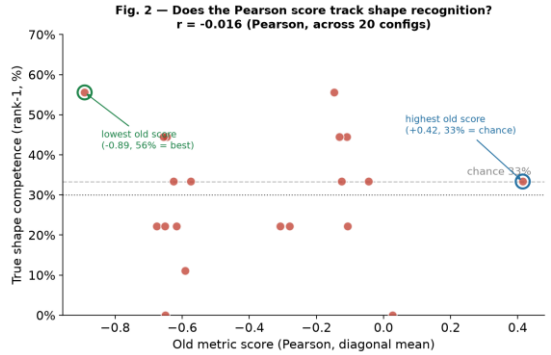
Code & data: github.com/pan112-sp/EGO

Task: a fixed 1000-LIF reservoir + plastic STDP readout maps visual shape → vocal-tract trajectory; evaluated by Pearson vs rank-1 (30 trials = 10 seeds × 3 shapes).

① The metric silently fails

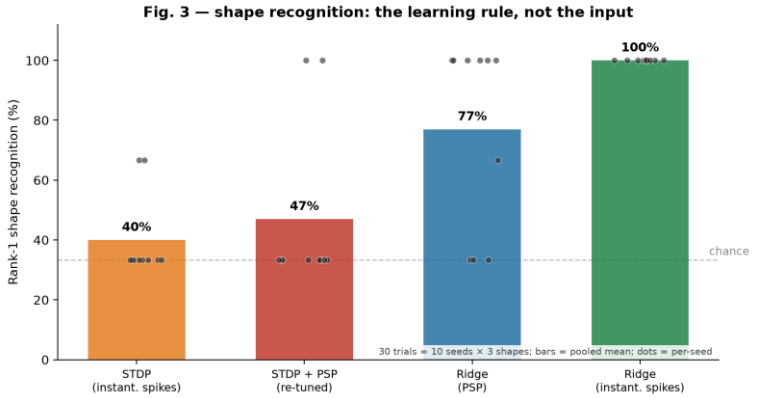


A "lazy" network (same output for every shape) scores HIGHEST — 0.902 — yet is at chance (33%) on rank-1.



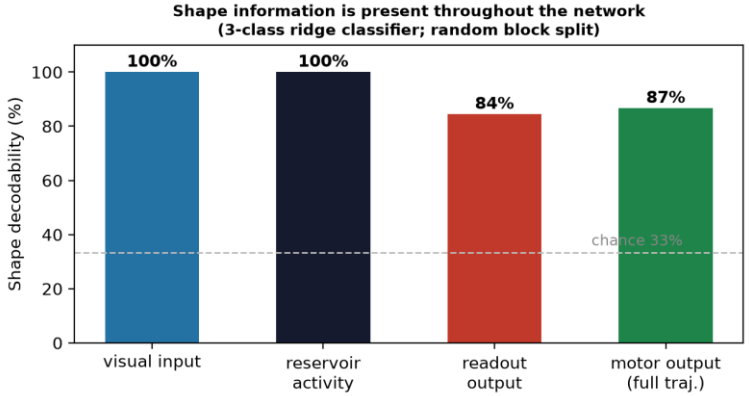
Across 20 configs: r = −0.016 (Pearson) between the old score and true shape recognition.

② The gap is real — the learning rule



Ridge is ONE readout on two inputs: PSP-filtered rates → 77%, the SAME instantaneous spikes STDP sees → 100%. NNLS (non-negative ridge) also 100% — non-negativity alone does not explain the gap. Under the tested configurations the local, online rule underperforms the global one.

③ Shape info decodable, mapping fails



3-class ridge classifier; within-run block split (10 train / 10 test blocks per shape). Within-run decodability — independent-trial generalization untested.

Correlation = output & target trends co-vary.
Mapping = output matches the CORRECT target for each input.
High correlation ≠ correct mapping.

Takeaway — a high correlation does NOT imply correct mapping; use metrics that directly measure the correct input→output correspondence.

Boundary — we do NOT claim STDP cannot learn in principle; only that it fails under the tested configurations and fair comparison.

Code: github.com/pan112-sp/EGO